

Introduction to machine learning

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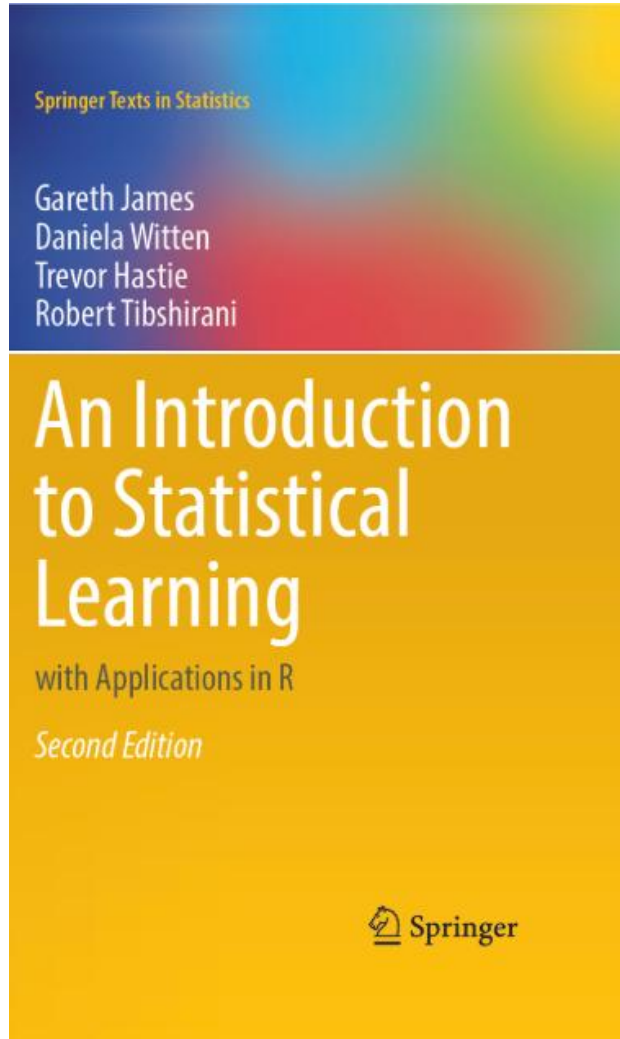


Course outline

Course outline

	Monday	Tuesday
Morning	Introduction	Deep learning
Afternoon	Tree based methods Support vector machines	Tutorial

Disclaimer



Most of the figures in this slides
are taken from this book, which is
available for free at:

<https://www.statlearning.com/>

Machine learning

Lecture 1 Introduction

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Background

Outline

- What is machine learning?
- Types of machine learning
- A timeline of machine learning
- Machine learning: Dr AI-skyll & Mr Hype
- Machine learning vs statistics

What is machine learning?

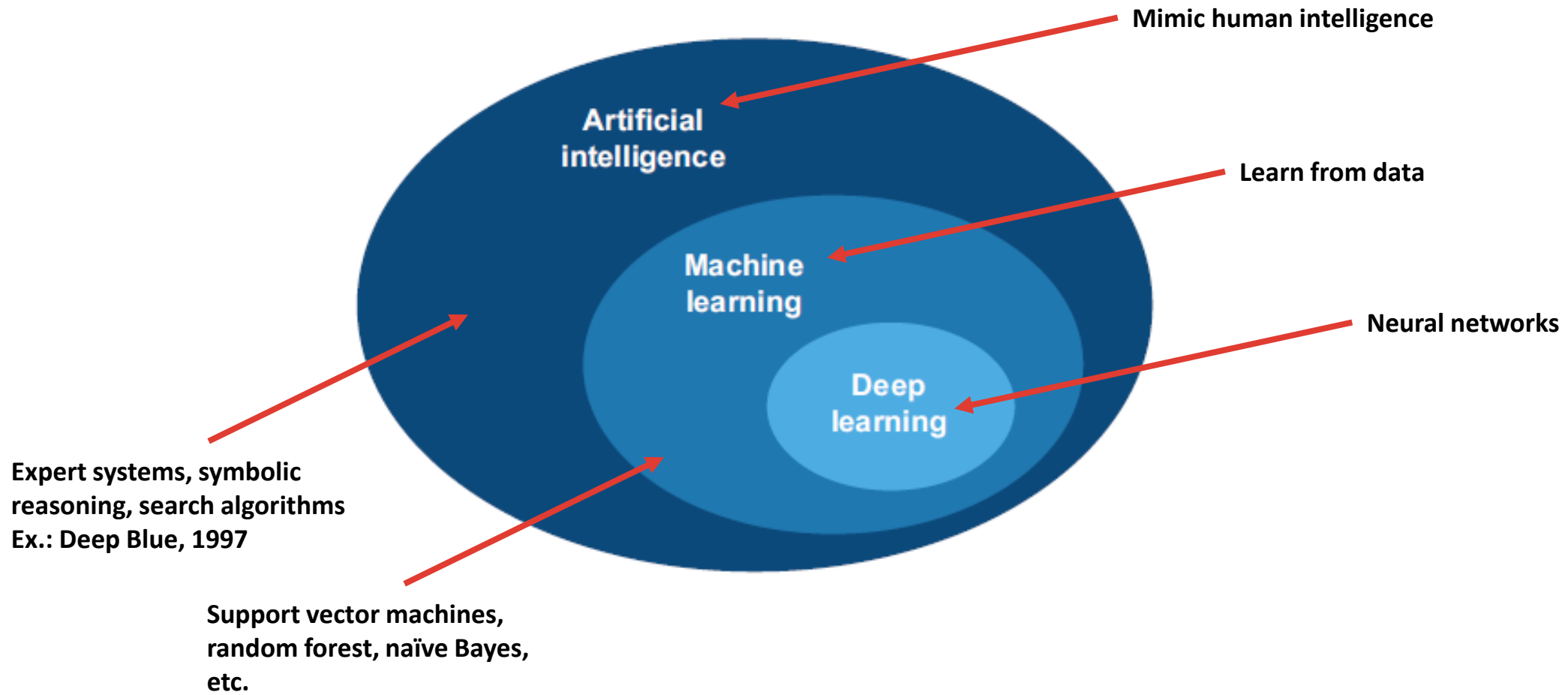
- The term was coined by **Arthur Samuels** in his 1959 paper titled *Some studies in machine learning using the game of checkers*
 - “a computer can be programmed so that it will **learn** to play”
 - “Programming computers to **learn** from experience should eventually eliminate the need for much of this detailed programming effort.”

What is machine learning?

Some definitions

- "Machine learning is the science of getting computers to act **without being explicitly programmed.**" – Andrew Ng (Stanford, Coursera)
- "Machine learning is a field of study in artificial intelligence concerned with the development and study of **statistical algorithms** that can **learn from data** and generalize to unseen data, and thus perform tasks without being explicitly programmed." – Wikipedia (29/05/2026)

What is machine learning?



source: Deep Learning with Python. F. Chollet

Types of machine learning

Three main machine learning “paradigms”

- Supervised learning
 - It requires a ground truth / labelled data
 - Regression (continuous outcome), classification (discrete outcome)
- Unsupervised learning
 - No labels
 - Task: find structure in data
 - Examples: clustering, dimensionality reduction (e.g. PCA)
- Reinforcement learning
 - Learning by trial and error
 - Maximising rewards
 - Examples: AlphaGo, RLHF

Types of machine learning

We will focus on supervised learning

- It requires a ground truth / labelled data
- Regression (continuous outcome), classification (discrete outcome)
- Special types of supervised learning
 - Semi-supervised learning
 - Only some samples have labels (often a minority)
 - E.g. use unlabelled data for clustering, labelled data to label clusters
 - Self-supervised learning
 - Don't require labels → the sample is the label
 - Examples: next-word prediction, autoencoders

A timeline of machine learning

- [1763] Discovery of Bayes theorem
- [1805] Least squares method proposed
- [1843] The differential engine
- [1913] Markov chains
- [1943] Artificial neuron
- [1950] Alan Turing's "Can machines think?"
- [1957] The perceptron (first trainable neural network)
- [1986] Backpropagation applied to train multi-layered neural networks
- [1995] Random forests
- [1995] Support vector machines
- [2006] "Deep learning"
- [2017] Transformers

Machine learning: Dr AI-skyll & Mr Hype

The success stories

Recommender systems:

- Netflix, Spotify, YouTube — billions of daily decisions
- Ad placement
- Social network feeds



Machine learning: Dr AI-skyll & Mr Hype

The success stories

Computer vision:

- 2012 ImageNet
- 2015 Superhuman image classification
- Medical imaging diagnostics / segmentation
- Content filtering
- Image / video generation



Machine learning: Dr AI-skyll & Mr Hype

The success stories

Science:

- Protein structure prediction (AlphaFold 2)
- Climate modelling
- Drug discovery
- Weather forecasting
- Mathematical reasoning

Machine learning: Dr AI-skyll & Mr Hype

The success stories

Natural language processing:

- Machine translation (Google Translate)
- Sentiment analysis, content filtering
- Large Language Models (LLMs)
 - Chatbots (ChatGPT, Claude)
 - Coding (Copilot, Claude Code, Codex)

Machine learning: Dr AI-skyll & Mr Hype

The success stories

Others:

- Medical diagnosis and early detection
- Fraud detection & credit scoring
- Anomaly detection in manufacturing
- Spam filtering
- Search ranking
- Churn & demand forecasting

Machine learning: Dr AI-skyll & Mr Hype

Don't Believe the Hype



Machine learning: Dr AI-skyll & Mr Hype

Don't Believe the Hype

- “Machines will be capable of doing any work a man can do within 20 years”
Herb Simon, 1965
- In 80s, expert systems were the future; they were abandoned in the 90s
- In 2016, IBM Watson promised to revolutionise oncology, but was found to make dangerous recommendations and later abandoned
- Since mid-2010s, multiple companies promised to reduce drug dev from 12 to 2-4 years: no AI-discovered drug has passed phase-III
- Since 2022, the advent of ChatGPT has led some people to predict the end of employment...
- The impact of ML is real... but often exaggerated.

Machine learning: Dr AI-skyll & Mr Hype

Don't Believe the Hype

- Logistic regression will beat other ML algorithms **most** of the time
- **Occam's Razor in ML:** when a simple model fits the data well, prefer it
- **Common factors behind ML success:** large labelled high-quality datasets, compute, well-defined objective, closed/simulated environment
- **Why linear models often win on tabular data:**
 - Low sample sizes → complex models overfit
 - High noise-to-signal ratio → flexibility is a liability
 - Features are already engineered
 - Interpretability requirements → regulators don't accept black boxes

Machine learning vs statistics

- Leo Breiman (2001), "Statistical Modeling: The Two Cultures",
 - Data modelling culture (statisticians):
 - assume a stochastic data-generating model;
 - estimate parameters;
 - inference is the goal
 - Algorithmic modelling culture (ML):
 - treat the data mechanism as unknown;
 - optimise predictive accuracy;
 - the black box is acceptable
- Breiman estimated ~98% of statisticians were in the data modelling culture at the time

Machine learning vs statistics

Is logistic regression machine learning?

- Used in both fields: the algorithm is the same, the goals differ
- Statistician:
 - wants coefficients, p-values, confidence intervals (inference)
- ML practitioner:
 - wants held-out AUC, calibration (prediction)

Machine learning vs statistics

Tibshirani's glossary



Machine learning	Statistics
network, graphs	model
weights	parameters
learning	fitting
generalization	test set performance
supervised learning	regression/classification
unsupervised learning	density estimation, clustering
large grant = \$1,000,000	large grant = \$50,000

“Machine learning is the art of producing misleading conclusions from huge datasets whereas statistics is the much more efficient science of doing so from small ones.”

Stephen Senn (statistician)

Machine learning vs statistics

- Some people think ML is a computer science fiefdom
- In fact, most ML algorithms were proposed by statisticians. Eg:
 - Random forests by Leo Breiman
 - Support-vector machines by Vladimir Vapnik
- Modern view: the statistics & ML cultures are converging:
 - causal ML, conformal prediction, uncertainty quantification

Fundaments

Outline

- Introduction to supervised learning
- Assessing model accuracy
- Resampling methods
- Model selection

Intro to supervised learning

Motivating example

- We are asked to investigate association between advertising and sales
- We are given a dataset with 4 variables

- sales
- TV
- radio
- newspaper

} We cannot directly change

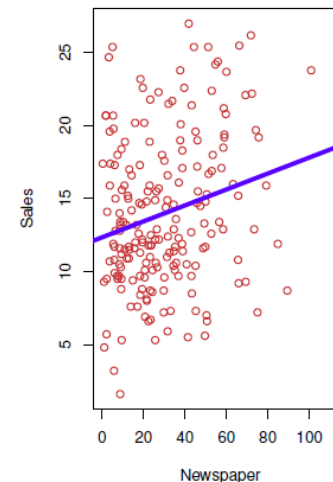
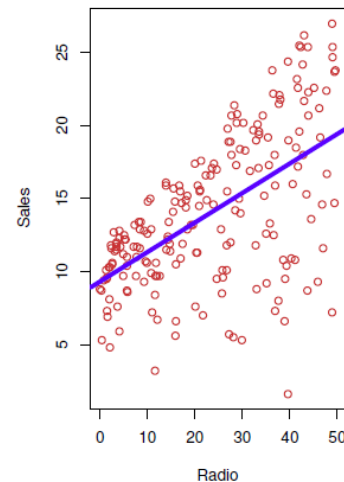
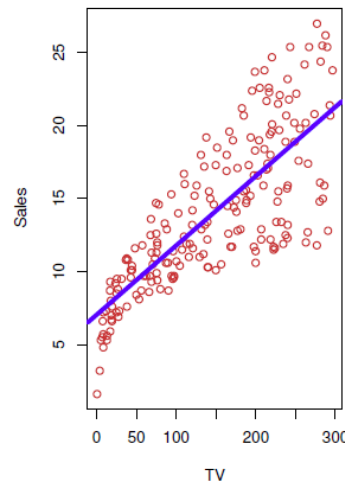
} We can change

Output variable

Outcome
Dependent variable
Response

Input variables

Predictors
Independent variables
Features



Intro to supervised learning

More generally, suppose that we observe:

- a quantitative response Y
- p different predictors $X = (X_1, X_2, \dots, X_p)$
- We assume there's a relationship between them:

$$Y = f(X) + \epsilon$$

Fixed but
unknown
function

Error term

- Supervised learning is the set of approaches for estimating f

Why estimate f ?

Two main reasons: prediction and inference

Prediction

- We have inputs X , we want to predict Y
- We calculate:

$$\hat{Y} = \hat{f}(X)$$

- where $\hat{f}$ is our estimate of f
- In prediction, $\hat{f}$ is often treated as a black box
- The accuracy of $\hat{Y}$ depends on:
 - *reducible error* ← Can be reduced with more data / better ML technique
 - *irreducible error* ← Cannot be reduced, because f is also a function of ϵ

Why estimate f ?

Two main reasons: prediction and inference

Inference

- We are interested in the association between Y and X
- $\hat{f}$ cannot be treated as a black box, we're interested in its form!
- We might be interested in the questions:
 - Which predictors are associated with the response?
 - What is the relationship between the response and each predictor?
 - Is the relationship between Y and X linear or not?

In ML, we are often interested in prediction, but interest for (causal) inference is growing

How do we estimate f ?

- Many different linear and non-linear approaches.
- They all need *training data*: a set of n observations / data points
 - Use these data to ‘train’ or ‘teach’ our method how to estimate f
 - Let x_{ij} denote the value of the j -th predictor for observation i
 - Let y_i denote the value of response variable for the i -th observation
 - Data: $\{(x_1, y_1), (x_2, y_2), \dots (x_n, y_n)\}$ where $x_i = (x_{i1}, x_{i2}, \dots x_{ip})^T$
- Our goal:
 - Find a function $\hat{f}$ such that $Y \approx \hat{f}(X)$ for any observation (X, Y)
- Two main groups:
 - parametric
 - non-parametric

How do we estimate f ?

Parametric methods

- First, make an assumption on functional form of f
 - e.g. f is linear in X

$$f(X) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p \quad \leftarrow \text{linear model}$$

Now, we “only”
have to estimate
 $p+1$ coefficients!

- Second, we need a procedure to *fit* or *train* model
 - in linear models, most common approach is *(ordinary) least squares*

Least squares is
only one of many
possible ways!

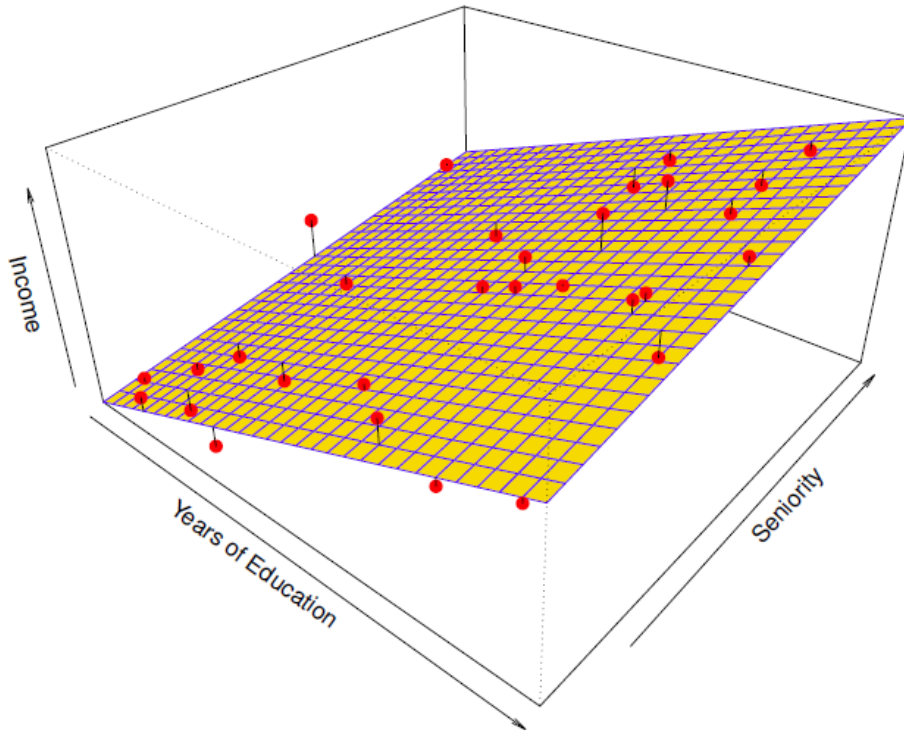
How do we estimate f ?

Parametric methods

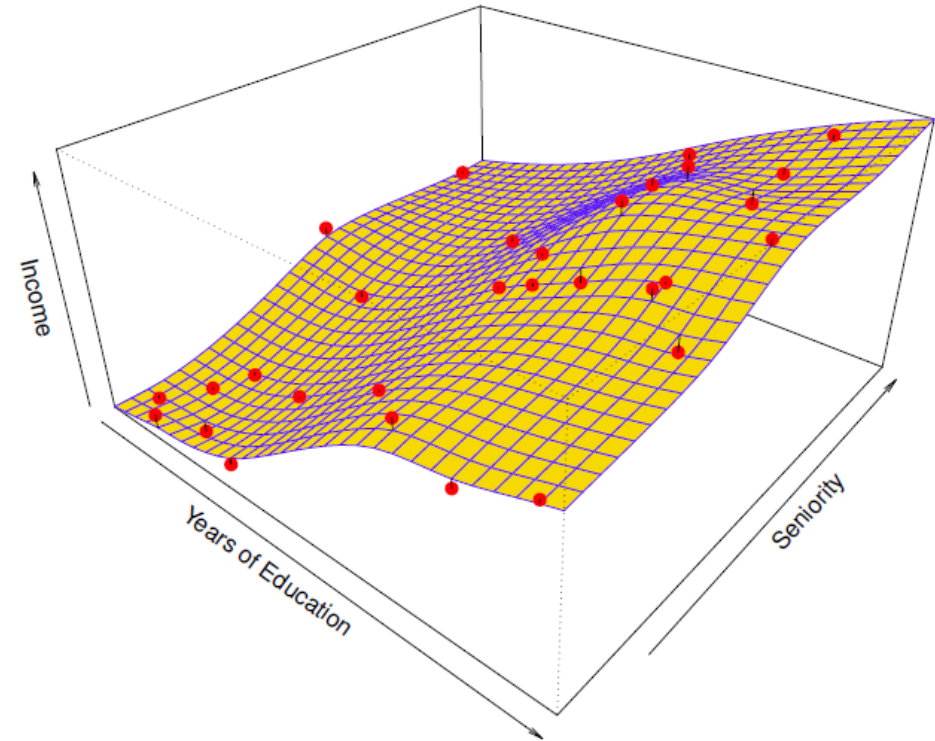
- Parametric methods simplifies the model fitting, but...
- Disadvantage: model chosen will usually not match form of f
 - if chosen model is far from f , the fit will be poor
- Potential solution: *flexible* models that can fit different functional forms
 - flexible models $\rightarrow$ more parameters $\rightarrow$ risk of *overfitting*  more on this later!

How do we estimate f ?

Parametric methods



Linear fit



**Non-parametric fit
(thin-plate spline)**

How do we estimate f ?

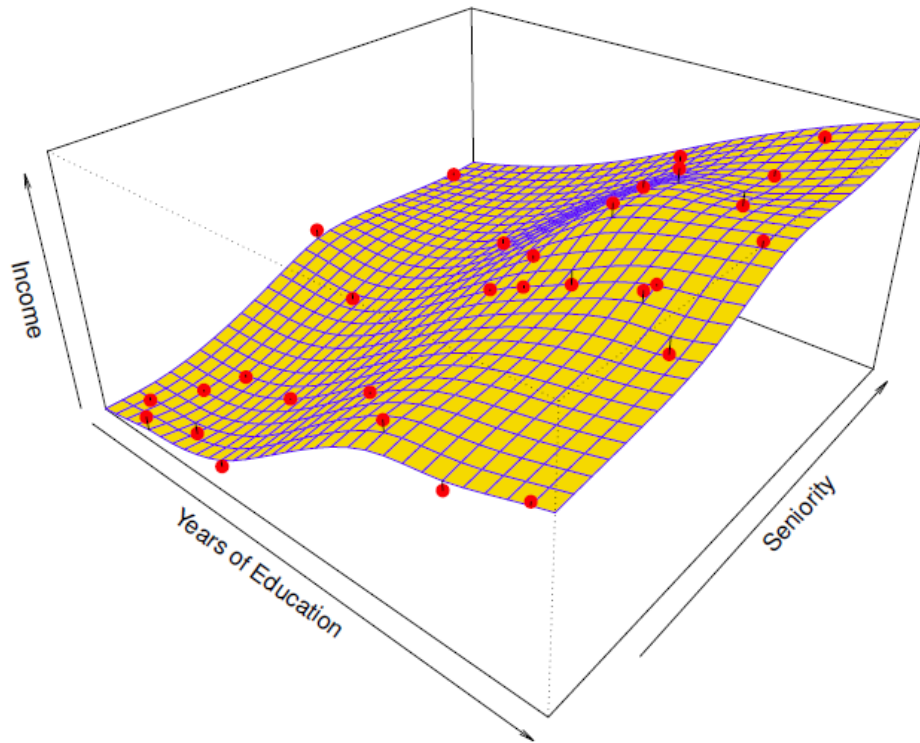
Non-parametric methods

Do not make assumptions on functional form of f

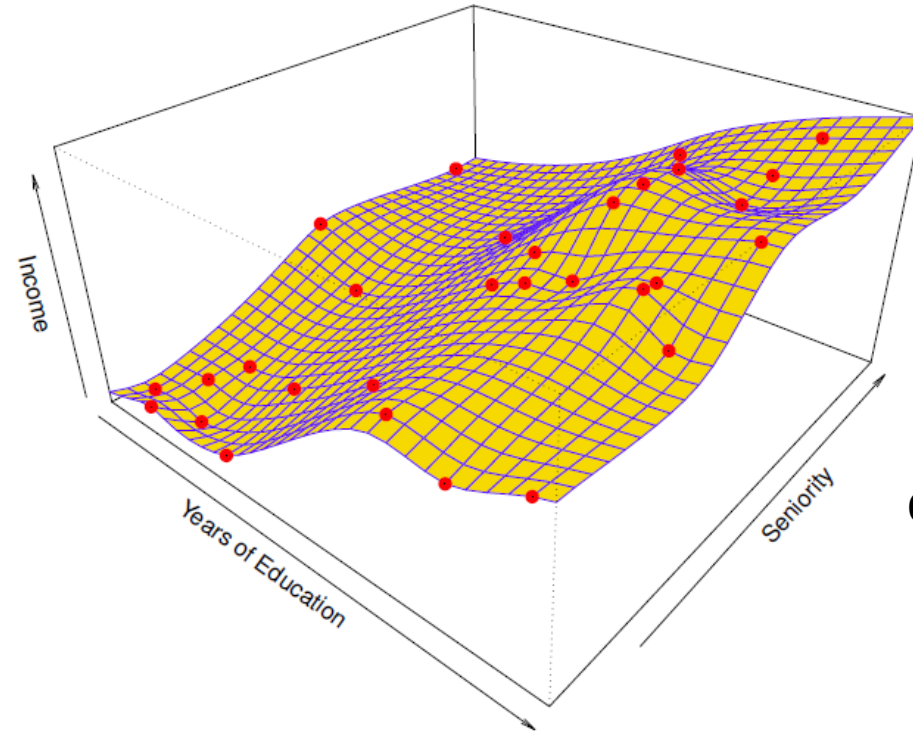
- can fit a wider range of shapes of f
- but, they require more data
- usually, they have a hyper-parameter to control flexibility
 - e.g. *smoothness* in splines

How do we estimate f ?

Non-parametric methods



**Thin-plate spline
(smooth)**



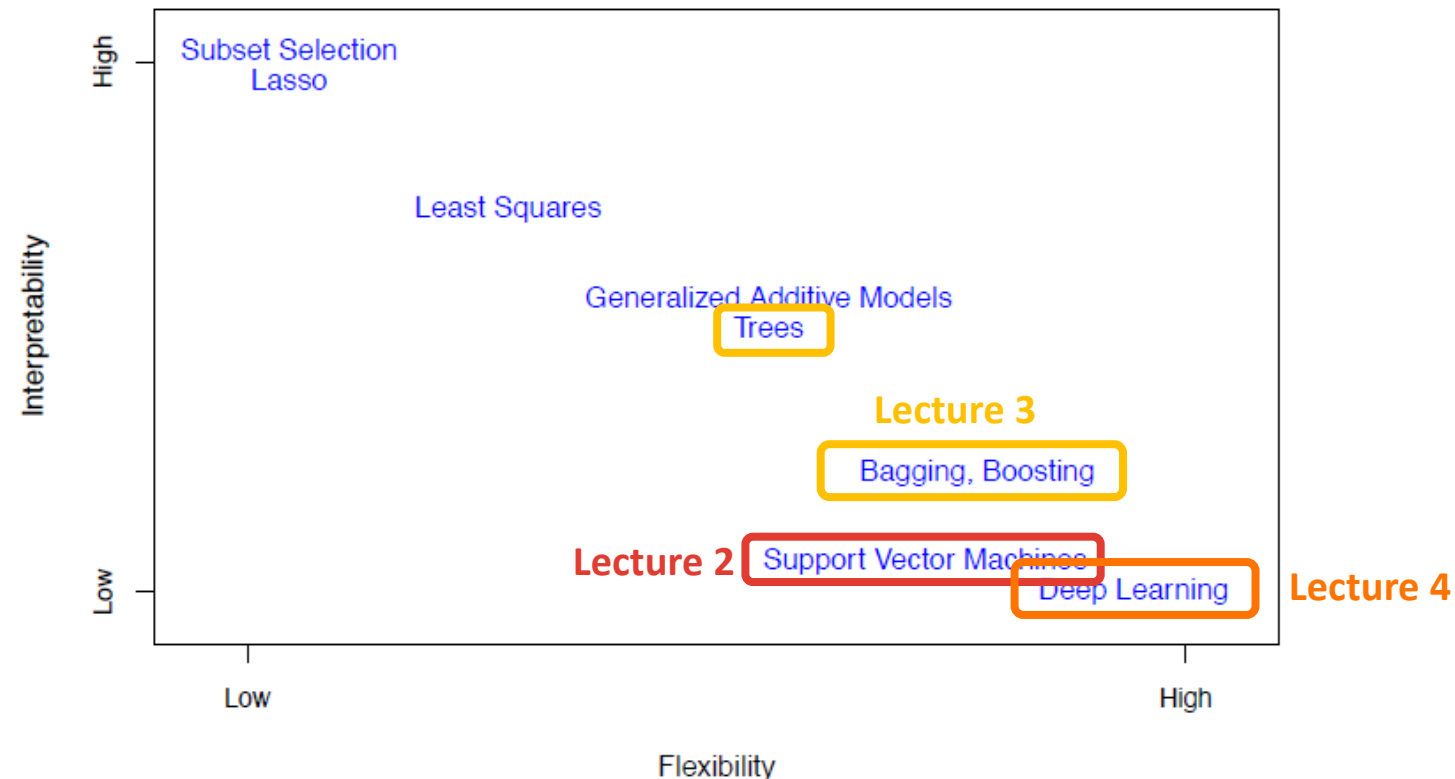
Overfitting

**Thin-plate spline
(rough)**

Trade-Off Between Accuracy and Interpretability

Why choose a more restrictive approach over a flexible approach?

- One reason is **interpretability**



Regression Versus Classification Problems

- Variables can be either
 - quantitative (numerical): age, height, income, price
 - qualitative (categorical): gender, marital status, cancer diagnosis
- Problems with:
 - quantitative response → regression
 - qualitative response → classification
 - exception: logistic regression (but it estimates probabilities, so...)
- We tend to choose ML methods based on nature of response
 - e.g.: linear regression for regression, logistic regression for classification
 - but: most ML methods can do both: e.g. random forest, SVMs, boosting

Outline

- Introduction to supervised learning
- **Assessing model accuracy**
- Resampling methods
- Model selection

Assessing model accuracy

- Why not just learn the best ML method?
- The **No free lunch** theorem in ML / optimisation
 - no one methods outperforms others in all datasets
- Hence, important to decide which method is best in a particular dataset

Assessing model accuracy

Measuring the Quality of Fit

- How well do model predictions match the observed data?
- In regression, mean squared error (MSE) is commonly used

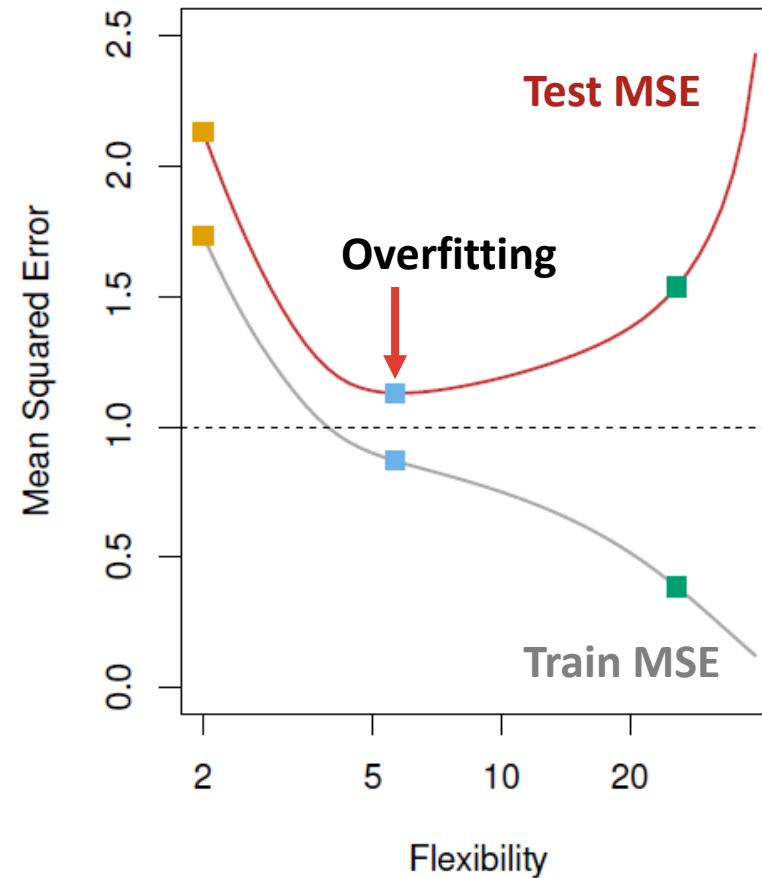
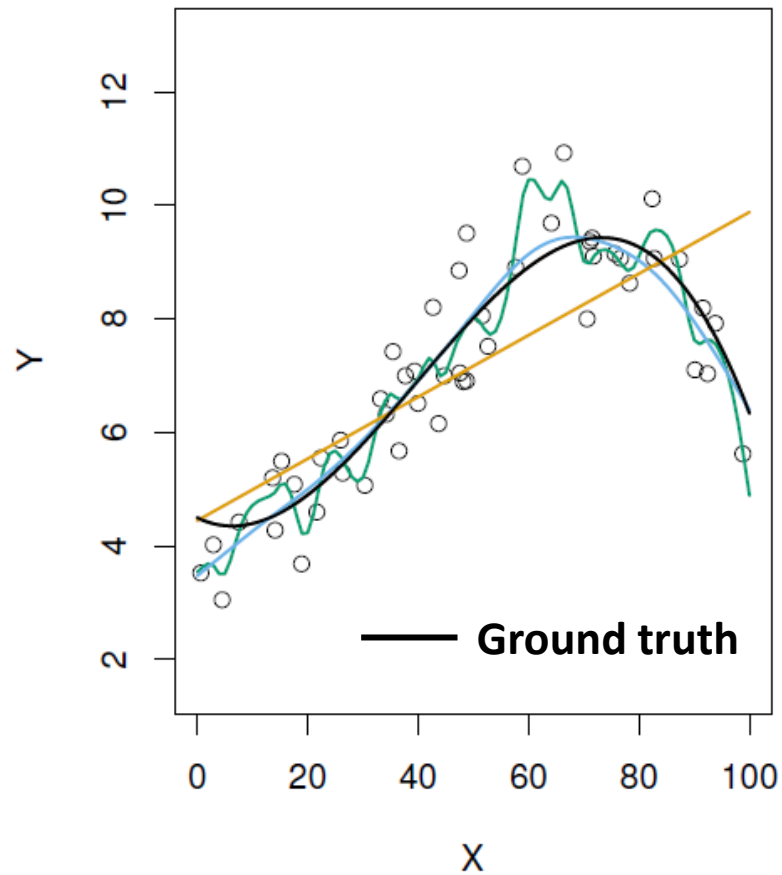
$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$


Which data should we use here?

- We are interested in performance on unseen data → test data / set
 - but what if we don't have a test set?
 - can't we just use the model with the best train MSE?

Assessing model accuracy

Measuring the Quality of Fit



Assessing model accuracy

The Bias-Variance Trade-Off

- Expected test MSE for a given value x_0 can be decomposed as:

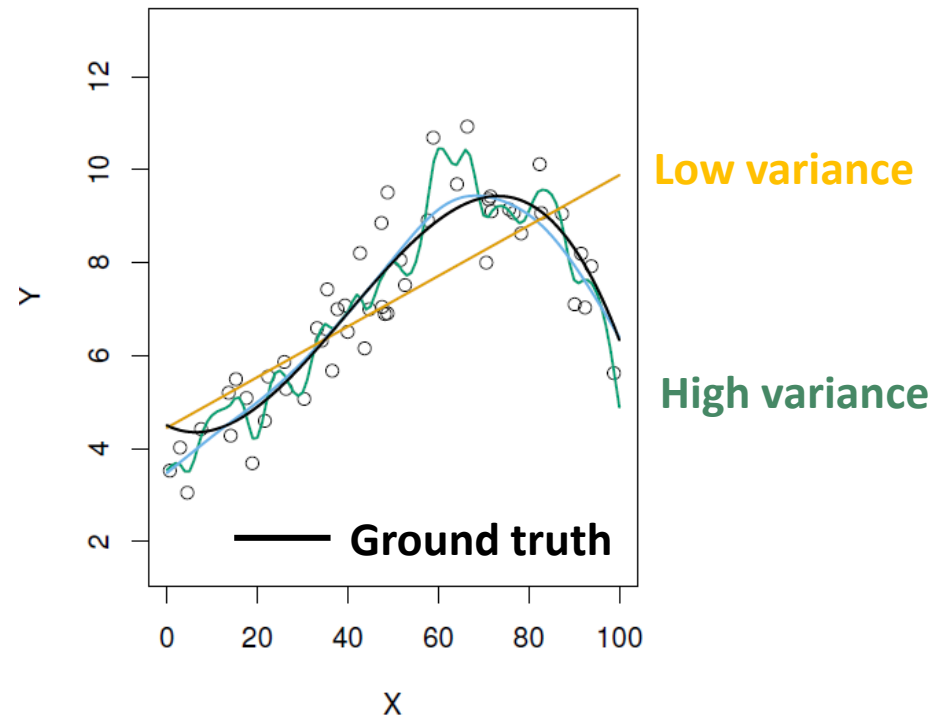
$$E \left(y_0 - \hat{f}(x_0) \right)^2 = \text{Var}(\hat{f}(x_0)) + [\text{Bias}(\hat{f}(x_0))]^2 + \text{Var}(\epsilon).$$

- Aim:** choose a method with low variance and low bias

Assessing model accuracy

The Bias-Variance Trade-Off

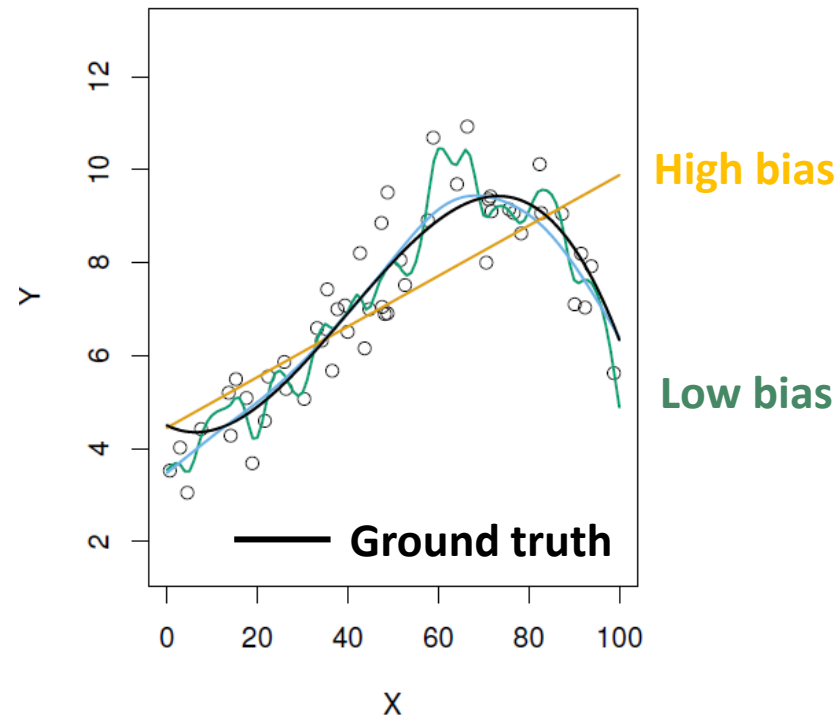
- What do we mean by variance of a statistical / ML method?
 - Amount by which $\hat{f}$ would change with a different training set



Assessing model accuracy

The Bias-Variance Trade-Off

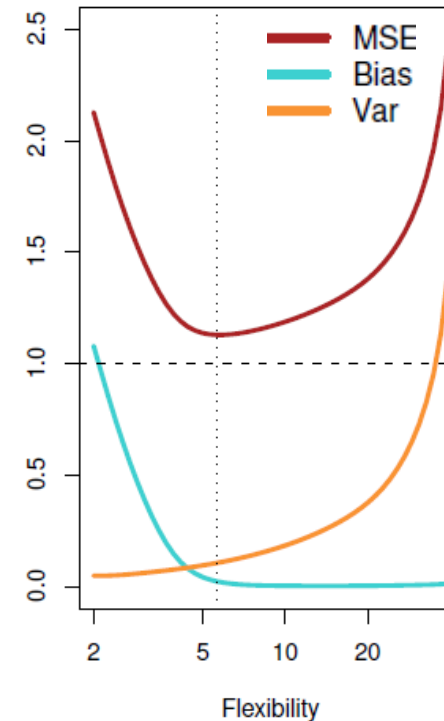
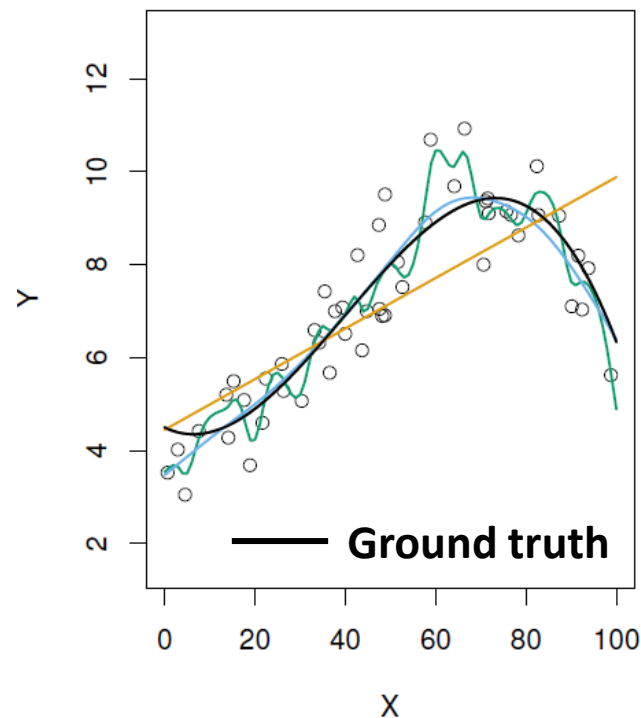
- What do we mean by bias of a statistical / ML method?
 - Error introduced by assumptions ← E.g. approximating a complicated real-life problem by a much simpler model



Assessing model accuracy

The Bias-Variance Trade-Off

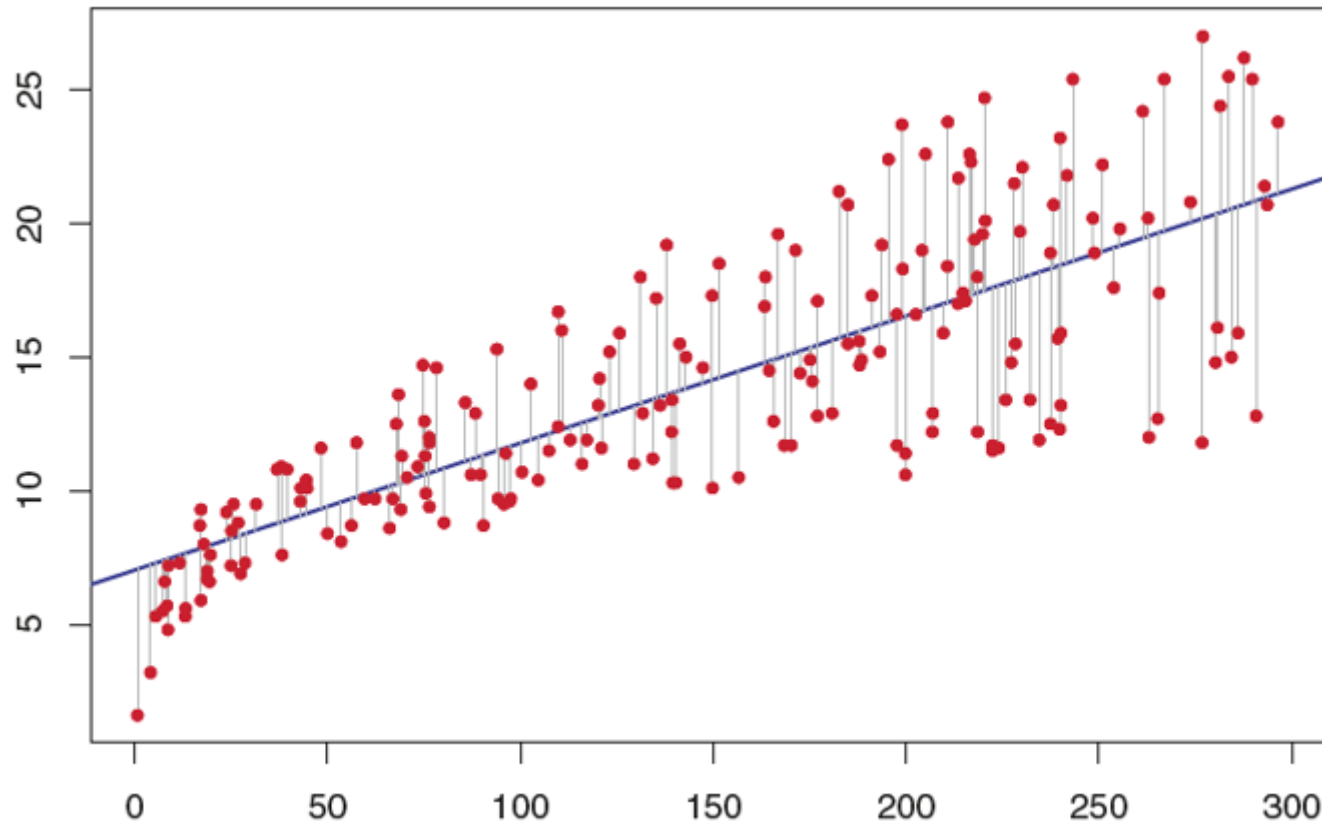
- In general, flexible models have high variance and low bias
- Test MSE is determined by relative change of bias and variance



**Bias-variance
trade off**

Assessing model accuracy

R-squared (R^2)



$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

% of explained variance

$$SS_{\text{res}} = \sum (y_i - f_i)^2$$

$$SS_{\text{tot}} = \sum_i (y_i - \bar{y})^2$$

Assessing model accuracy

The Classification Setting

- So far, we focused on regression
- Most of it applies to classification, but
- A common approach to quantifying accuracy:
 - Error rate

$$\frac{1}{n} \sum_{i=1}^n I(y_i \neq \hat{y}_i)$$

Indicator variable

Predicted class

- The best classifier → smallest test error rate
- A common metric is also accuracy, defined as $1 - error_rate$

Assessing model accuracy

The Classification Setting

- However, error rate / accuracy have shortcomings
- **Example:** we built a model to predict which customers will default

		<i>True default status</i>		
		No	Yes	Total
<i>Predicted default status</i>	No	9644	252	9896
	Yes	23	81	104
	Total	9667	333	10000

Error rate: 2.75%

Is that any good?

Note: A classifier that always predicts 'No' will have an error rate of 3.33%

Only 81 out of 333 (24.3%) who actually defaulted were correctly identified!

Assessing model accuracy

The Classification Setting

The confusion matrix

		<i>True class</i>		
		– or Null	+ or Non-null	Total
<i>Predicted class</i>	– or Null	True Neg. (TN)	False Neg. (FN)	N*
	+ or Non-null	False Pos. (FP)	True Pos. (TP)	P*
	Total	N	P	

Sensitivity or recall: $\frac{TP}{P}$ **Specificity:** $\frac{TN}{N}$ **Precision or PPV:** $\frac{TP}{P^*}$

More trade-offs in ML: sensitivity vs specificity, precision vs recall

$$F_1 = 2 \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

Assessing model accuracy

The Classification Setting

- ML models generally output a probability, not class predictions
- We transform probabilities into class predictions with a threshold. E.g.:

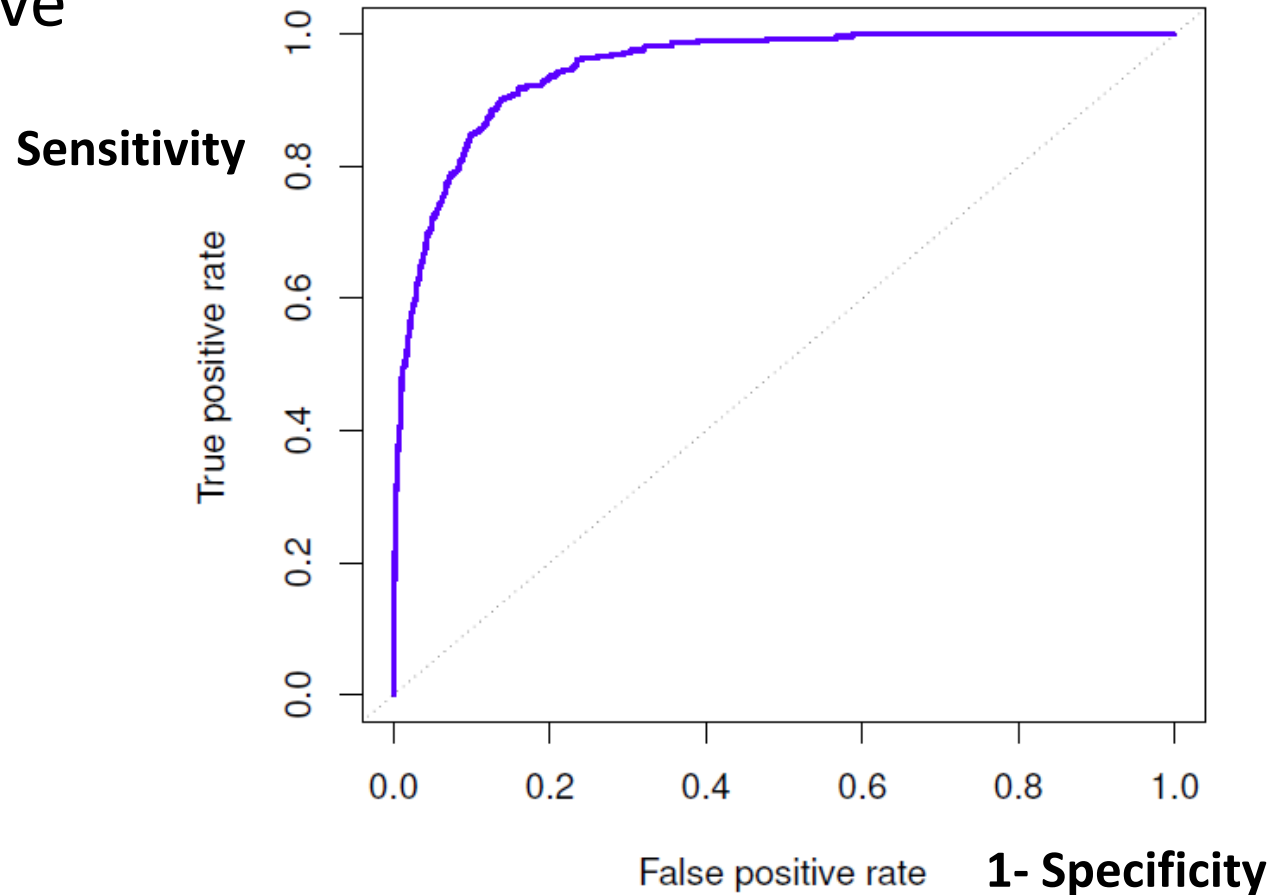
$$\Pr(\text{default} = \text{Yes} | X = x) > 0.5$$

- Playing with threshold, we shift trade-off between sensitivity & specificity
 - Lower threshold $\rightarrow$ + positives $\rightarrow$ + sensitivity / - specificity
 - Higher threshold $\rightarrow$ + negatives $\rightarrow$ - sensitivity / + specificity
- What if we could plot sensitivity & specificity pairs for all thresholds?

Assessing model accuracy

The Classification Setting

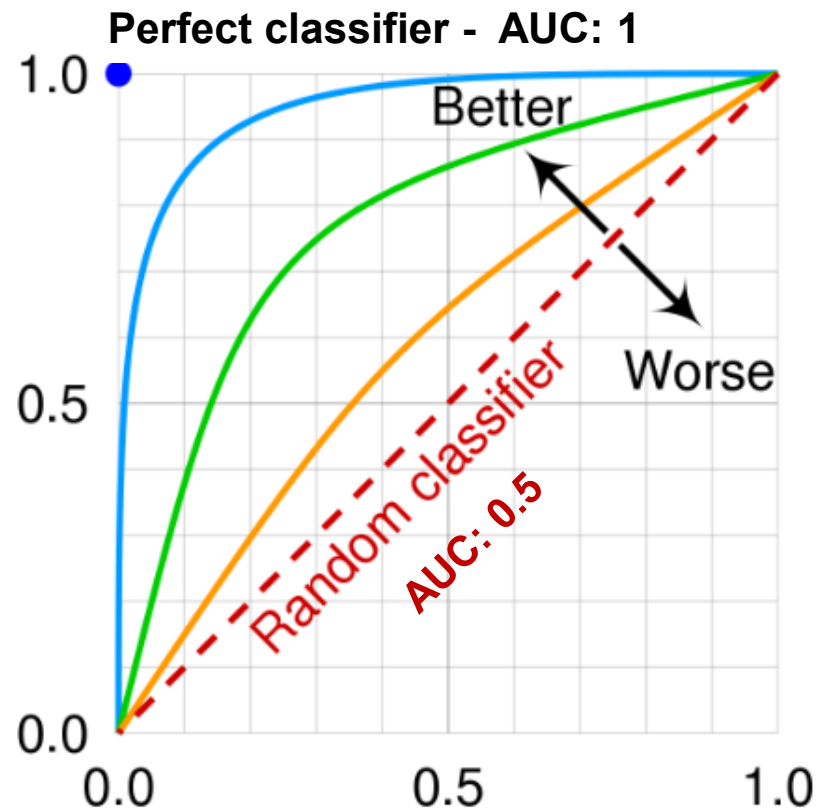
The ROC Curve



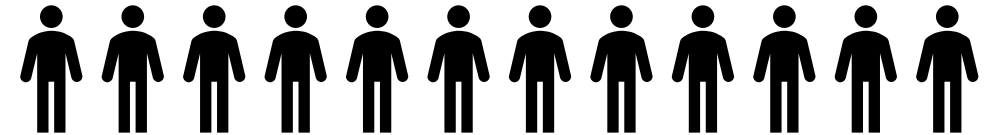
Assessing model accuracy

The Classification Setting

The ROC Curve – Area under the curve (AUC)



Ranked by
predicted risk



AUC: 0.5



AUC: 0.75



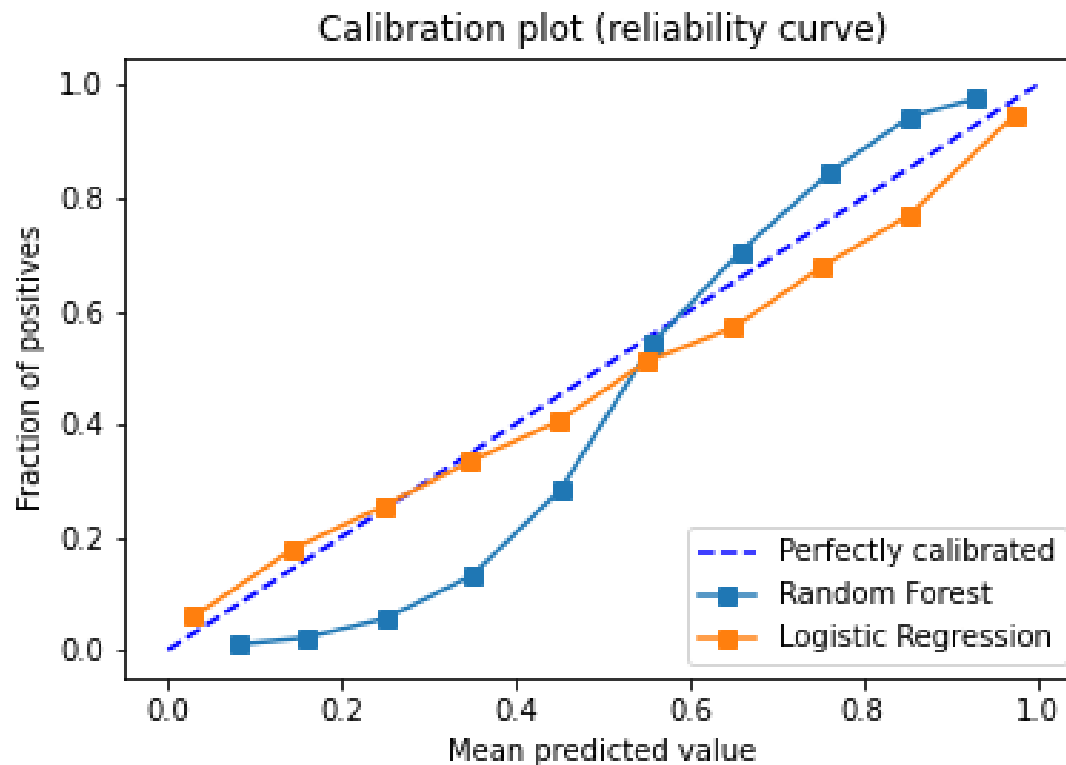
AUC: 1



Assessing model accuracy

The Classification Setting

Calibration



Calibration-at-large

- Intercept should be 0
- Slope should be 1

Assessing model accuracy

The Classification Setting

Brier score

$$BS = \frac{1}{N} \sum_{t=1}^N (f_t - o_t)^2$$

predicted probability

observed outcome

<u>Perfect classifier:</u>	BS = 0
<u>Random classifier:</u>	BS = 0.25
<u>Worst possible score:</u>	BS = 1

R squared R^2 [0,1]: explained variance in the outcome

Outline

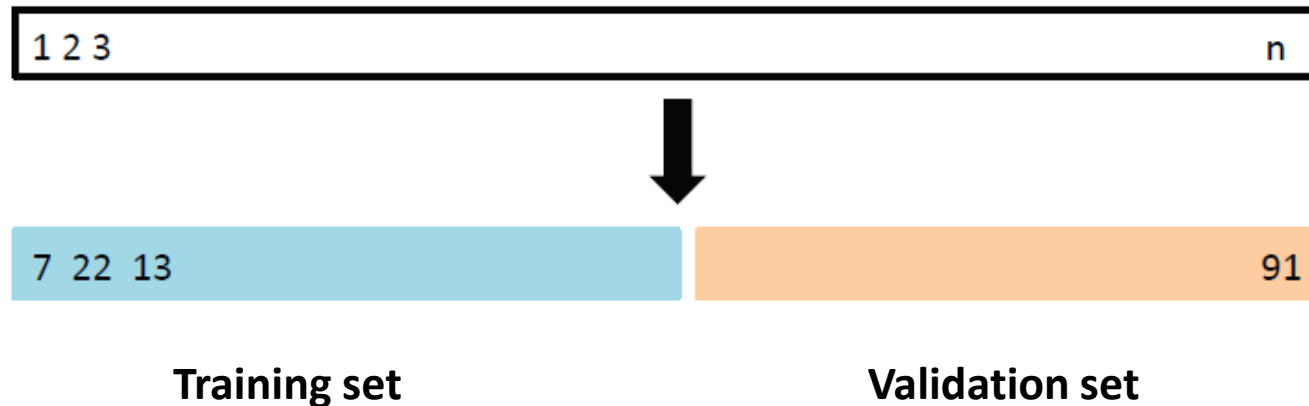
- Introduction to supervised learning
- Assessing model accuracy
- **Resampling methods**
- Model selection

- Repeatedly draw samples from dataset, refit model, obtain information
 - e.g. variability / robustness of model fit
- Here we discuss most common:
 - Cross-validation: Can be used for
 - assessment of model performance
 - model selection, more specifically hyperparameter tuning
 - Bootstrapping:
 - provides a measure of uncertainty of parameter or ML method performance

- Test error rate vs training error rate
- How can we estimate test error rate without a test set?
 - Mathematical adjustment to training error
 - Hold out subset of training set from model fitting:
 - The Validation Set Approach
 - Leave-One-Out Cross-Validation (LOOCV)
 - k-Fold Cross-Validation

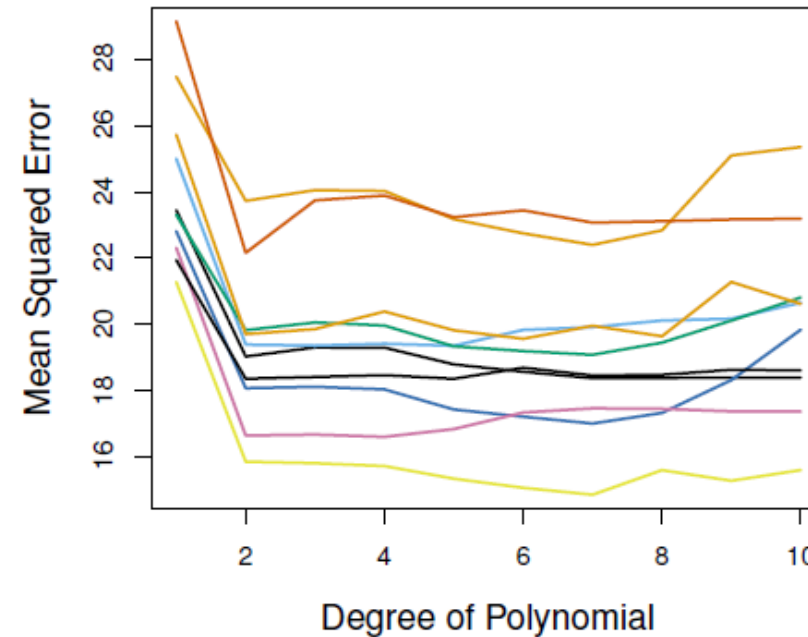
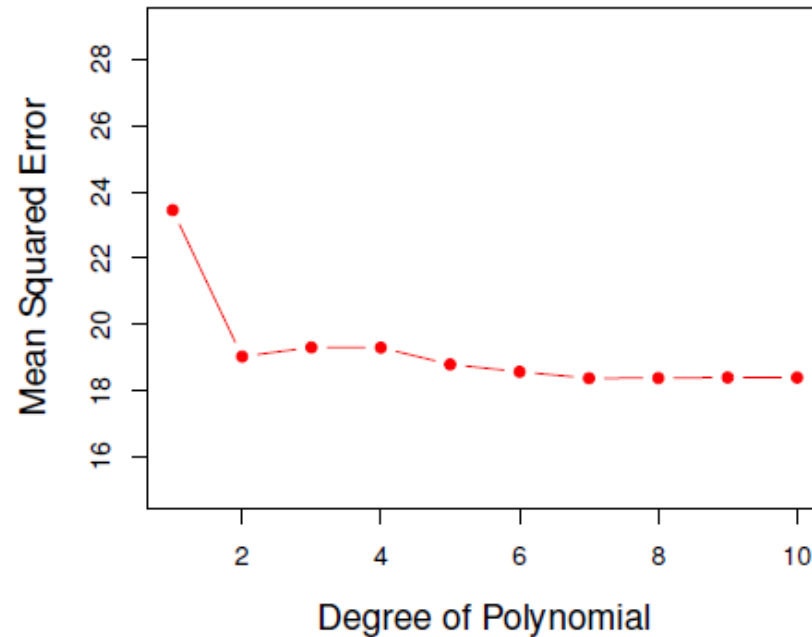
The Validation Set Approach

- Shuffle & split the data into training and validation (or hold-out) sets
- Fit model to training set
- Estimate performance on validation set



The Validation Set Approach

- Simple approach, but it suffers from high variability!

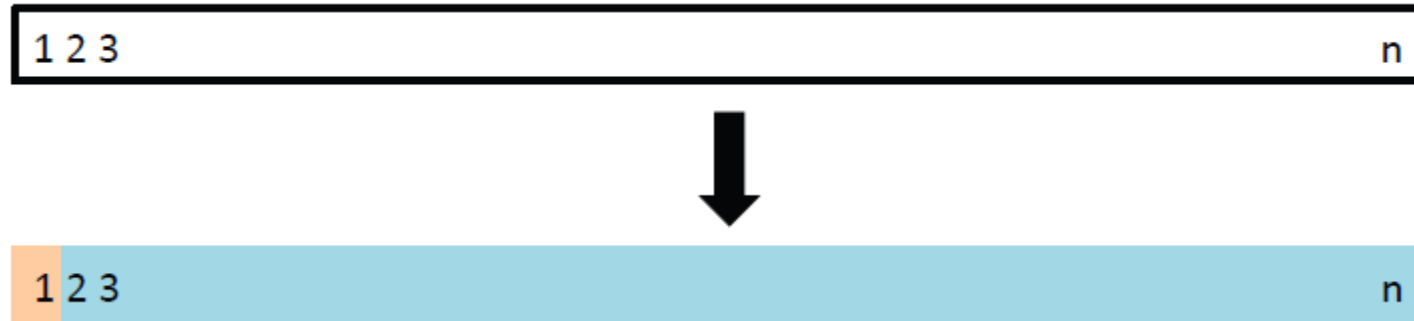


**Validation set
approach repeated
with different splits**

- We're also fitting the model to less data → overestimate test error

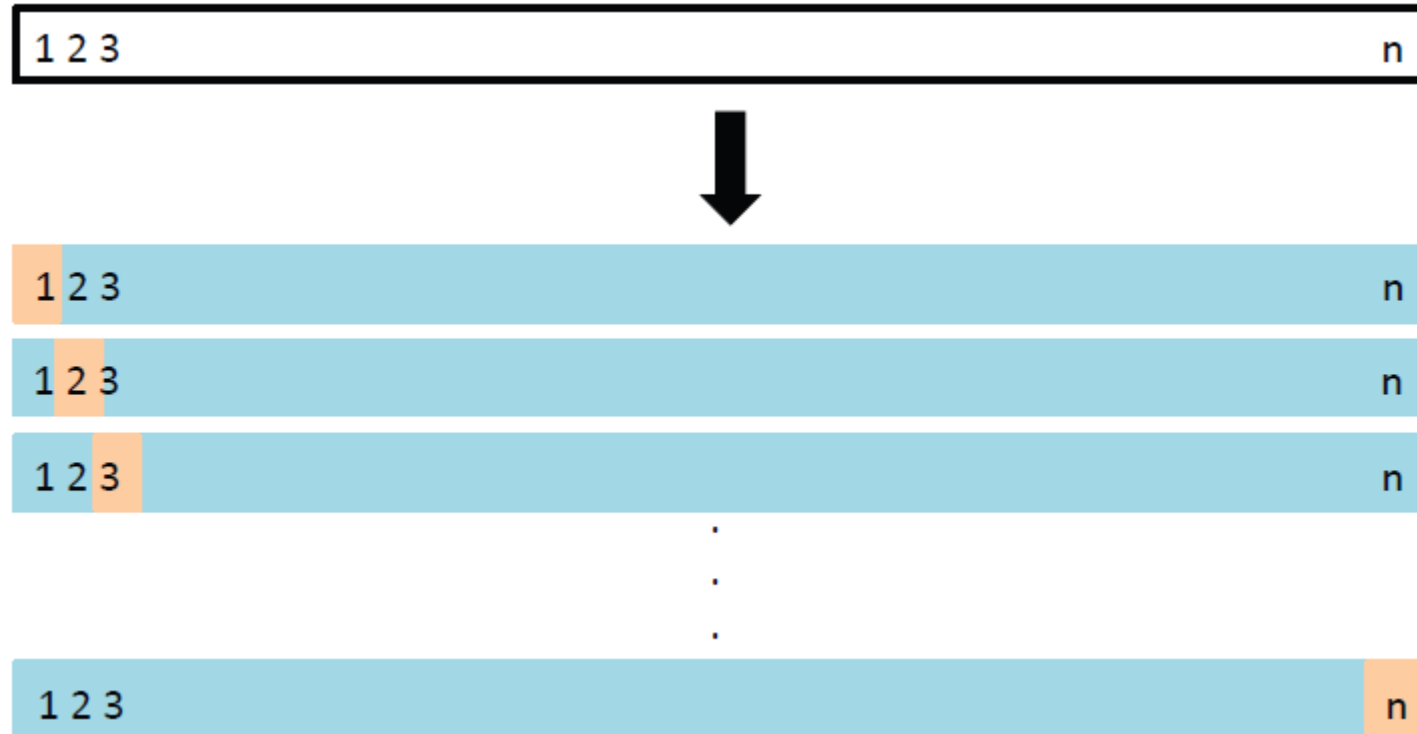
Leave-one-out cross validation (LOOCV)

- Again, split the data into training and validation sets, but...
- ...validation set is a single sample!



- It provides an unbiased estimator of test error, but highly variable!
- To reduce variance, repeat procedure using each sample as validation

Leave-one-out cross validation (LOOCV)



$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^n \text{MSE}_i.$$

Leave-one-out cross validation (LOOCV)

Advantages over the validation set approach

- ✓ Less bias: we use almost entire dataset
- ✓ Less variance: it always gives same result

Main disadvantage

- Computationally expensive: we fit the model n times

Maybe we can do
something in-between?

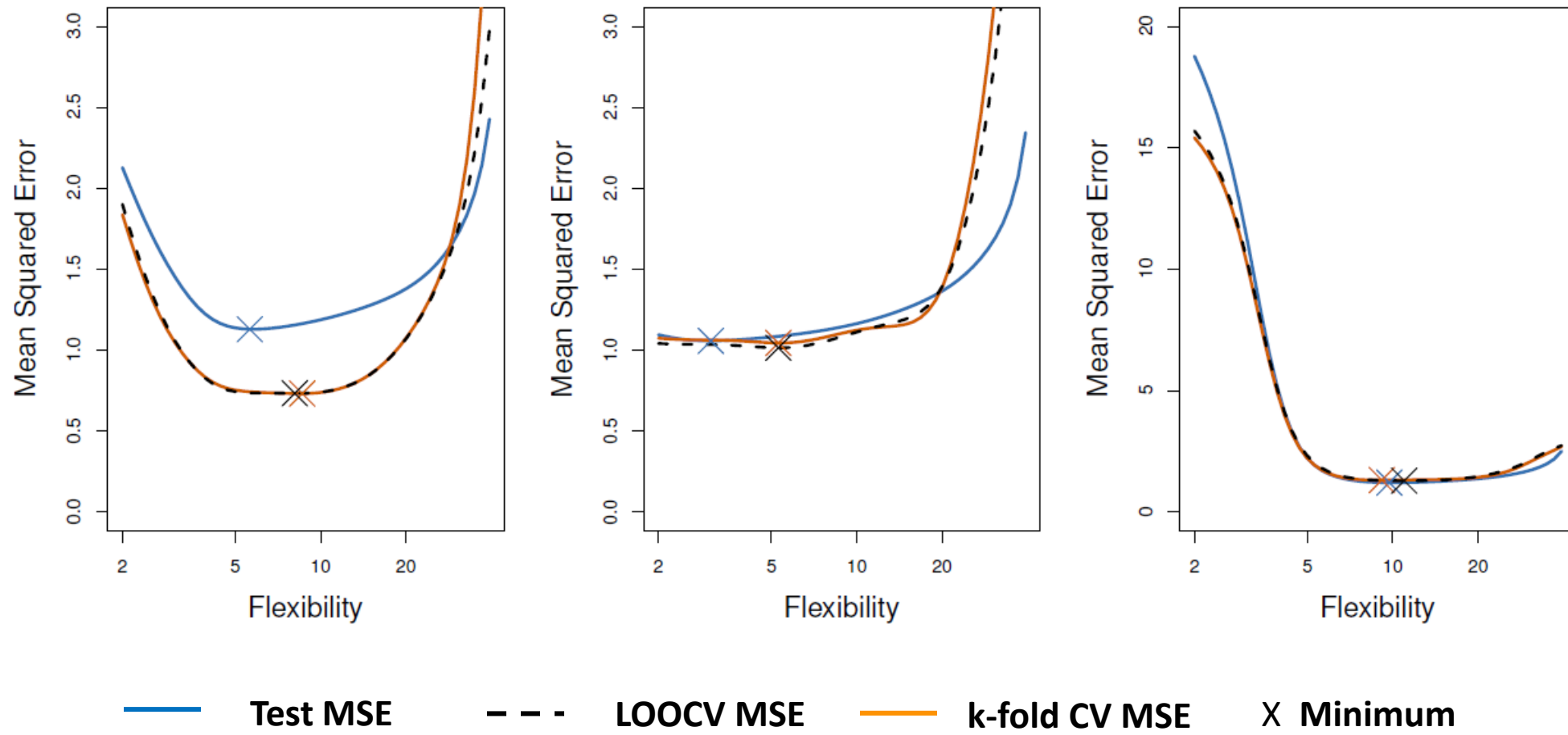
k-fold cross-validation

- Shuffle and split data into k equally sized folds
- Apply “LOOCV” to the folds

$$CV_{(k)} = \frac{1}{k} \sum_{i=1}^k MSE_i$$

	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Split 1	Test	Training	Training	Training	Training
Split 2	Training	Test	Training	Training	Training
Split 3	Training	Training	Test	Training	Training
Split 4	Training	Training	Training	Test	Training
Split 5	Training	Training	Training	Training	Test

k-fold cross-validation



k-fold cross-validation

Advantages of k-fold CV over LOOCV

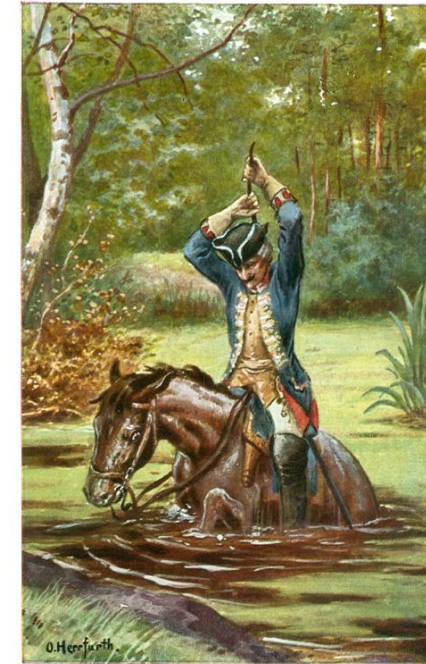
- Computationally much cheaper
 - fit model k times instead of n times
- It has slightly higher bias but lower variance
 - the performance estimates in LOOCV are highly correlated
 - the average of highly correlated outputs has high variance
 - in k-fold CV, k regulates the bias-variance trade-off
 - because of this trade-off, $k=5$ and $k=10$ are most commonly used

Bootstrapping

- It helps quantify uncertainty associated with an estimator / parameter



“Pulling oneself from the bootstraps” – meaning self-starting

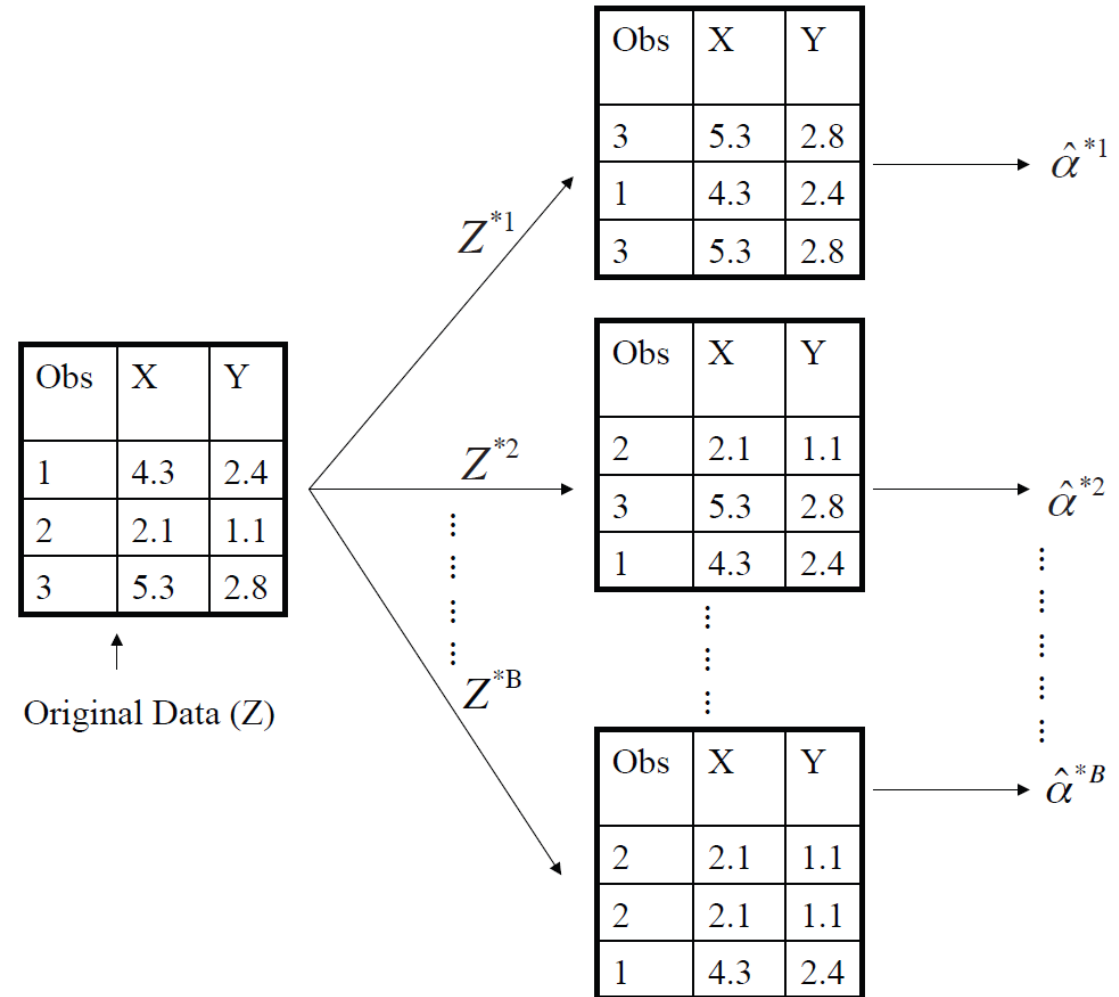


Baron Munchausen pulls himself (and his horse) out of a swamp by his own hair

Bootstrapping

How does it work?

- We generate B datasets (bootstraps) of n samples, sampling from our dataset with replacement
- We fit models to each bootstrap
- We aggregate performance estimates



How does it work?

1. Develop model using entire dataset (size n) and determine *apparent performance*
2. Repeat B (e.g. 100) times:
 - a. Generate *bootstrap sample* sampling n individuals with replacement
 - b. Develop model using bootstrap sample
 - c. Determine performance on bootstrap sample (*bootstrap performance*)
 - d. Determine performance on entire dataset (*test performance*)
 - e. Calculate optimism = bootstrap performance – test performance
3. Calculate:
 optimism adjusted performance = apparent performance – average(optimism)

Outline

- Introduction to supervised learning
- Assessing model accuracy
- Resampling methods
- **Model selection**

Model selection

- We've covered hyperparameter tuning using CV
- There are other aspects of a model we can select, namely
 - Feature selection
 - Model structure*
 - Dimensionality reduction

Feature selection (aka variable selection)

Motivation

- Including all variables in dataset in the model might not be optimal
- We might want to use variables not included in the dataset:
 - variables with different functional forms (e.g. polynomials)
 - interaction terms between variables
 - engineered features (e.g. BMI)
- The higher the p :
 - the higher the variance of the model (risk of overfitting)
 - the less interpretable the model will be
 - the more difficult the model will be (to use and to run)

Feature selection (aka variable selection)

Approaches

- Filter methods based on statistical measures (e.g. correlation)
 - Eliminating correlated predictors
 - Univariate selection: based on association with outcome
- Best subset selection: it quickly becomes unaffordable when p is high
- Search algorithms / wrapper methods:
 - Stepwise forward selection
 - Start with empty model, add 'best' predictor at each step
 - Stepwise backward selection / recursive feature elimination
 - Start with full model, remove 'worst' predictor at each step
- Shrinkage: using regularisation, shrinking coefficients to 0 (e.g. LASSO)

Dimensionality reduction methods

- So far, we saw approaches for select existing (or engineered) predictors
- Now, approaches to *transform* predictors and fit a model using them
- Let $Z_1, Z_2, \dots, Z_M$ represent $M < p$ linear combinations of our predictors

$$Z_m = \sum_{j=1}^p \phi_{jm} X_j$$

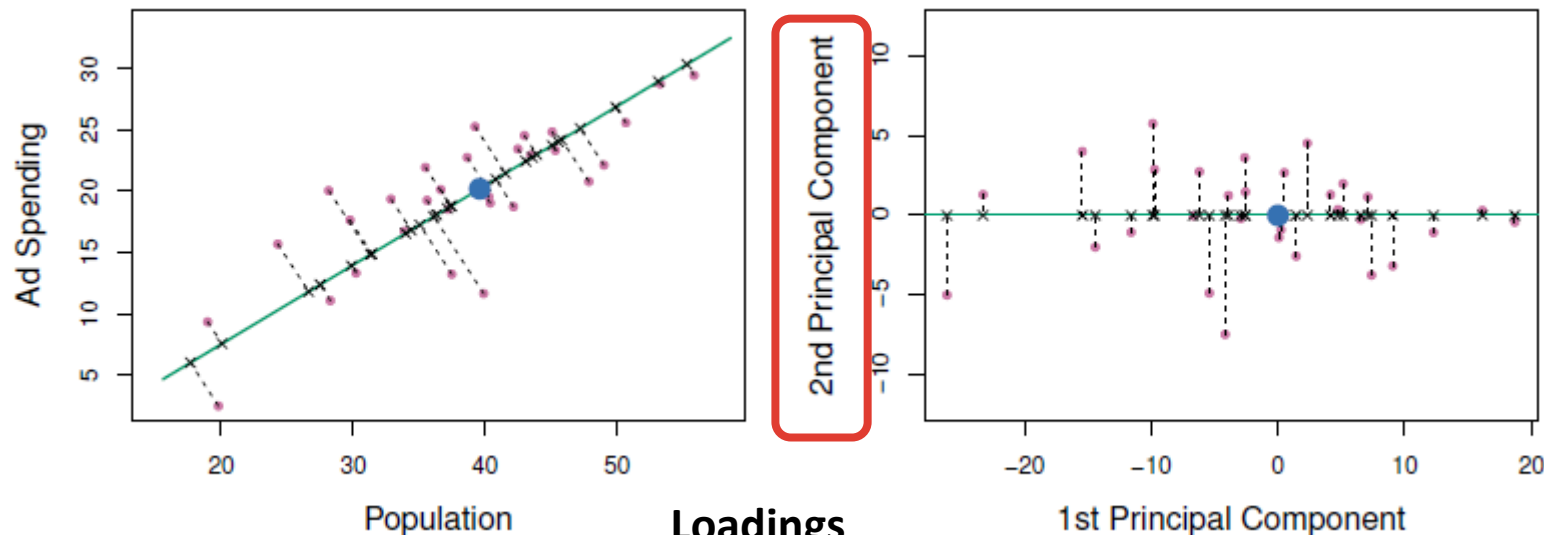
- We can fit a model (e.g. linear model):

$$y_i = \theta_0 + \sum_{m=1}^M \theta_m z_{im} + \epsilon_i, \quad i = 1, \dots, n,$$

Dimensionality reduction methods

Principal components analysis (PCA)

- PCA is a popular approach for deriving a low-dimensional set of variables
- The first PC is a vector in which data varies the most



Second PC is perpendicular to first

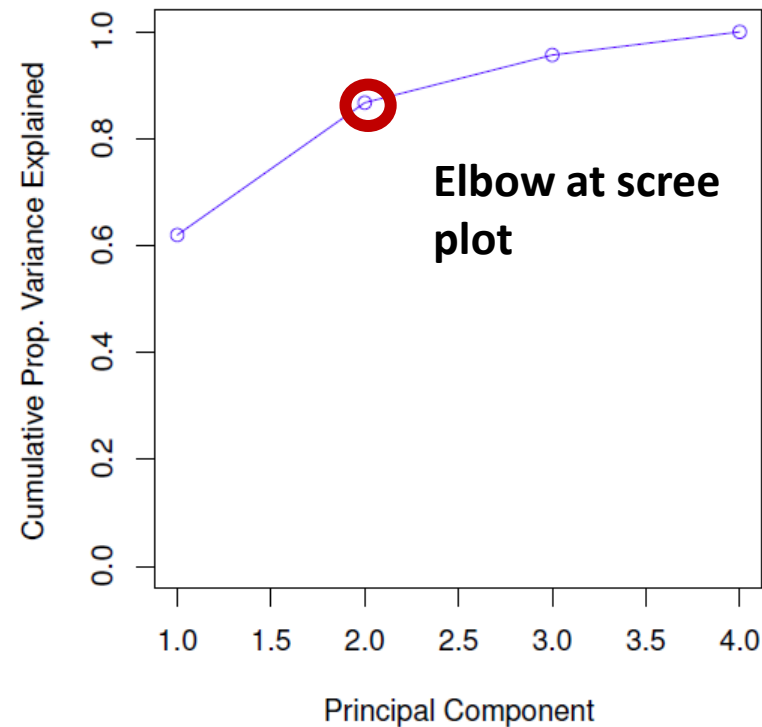
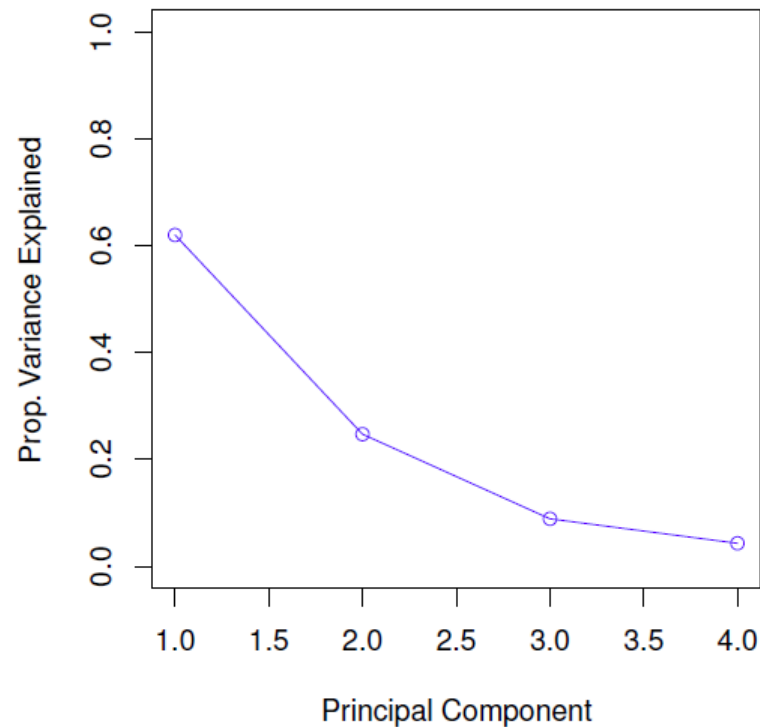
$$Z_1 = 0.839 \times (\text{pop} - \overline{\text{pop}}) + 0.544 \times (\text{ad} - \overline{\text{ad}}).$$

Principal component scores: $z_{i1} = 0.839 \times (\text{pop}_i - \overline{\text{pop}}) + 0.544 \times (\text{ad}_i - \overline{\text{ad}}).$

Dimensionality reduction methods

Principal components analysis (PCA)

- How many PCs should we fit?



Dimensionality reduction methods

Principal components regression (PCR)

- Fit regression model using principal components
- We assume the directions in which X varies most are associated with Y

